

Chapter 1: Exploring the Investigative World of Science

1. Why is one side of a puri thinner than the other?

Scientific Explanation:

- a. **Steam Expansion & Dough Layers:** When a rolled dough circle is dropped into hot oil, the water present in the dough turns rapidly into steam. As steam expands, it pushes the upper and lower layers of dough apart, inflating the puri like a balloon.
- b. **Gravity & Oil Submersion:** When dropped into oil, the bottom side remains in direct contact with the hot oil longer and gets fried/cooked faster, hardening its structure first. The top side expands upward with the rising steam. Because the expanding steam stretches the softer, unhardened top layer, that side becomes thinner compared to the bottom layer.

2. Are there more grains of sand on all the beaches and deserts of the world, or more stars in our galaxy?

Scientific Answer:

- a. **Stars in our Galaxy (Milky Way):** There are estimated to be between 100 billion to 400 billion stars (10^{11} to 4×10^{11}) in the Milky Way galaxy.
- b. **Sand Grains on Earth:** The estimated number of sand grains across all beaches and deserts on Earth is around 7.5×10^{18} (7.5 quintillion).
- c. **Comparison:** There are vastly more grains of sand on Earth than stars in our single galaxy (Milky Way). However, if we compare sand grains on Earth to the total number of stars in the observable universe (estimated at 10^{22} to 10^{24} stars across trillions of galaxies), there are far more stars in the universe than grains of sand on Earth!

3. Right from Grade 6, we've observed the incredible diversity of plants and animals around us. From the different shapes of leaves to the many kinds of insects—why has nature created such a vast variety?

Scientific Answer:

- a. **Adaptation to Environments:** Different living organisms live in different habitats (deserts, aquatic regions, mountains, forests). Diverse leaf shapes, body structures, and

traits evolved to help organisms adapt to their specific climate, light availability, and water conditions.

- b. **Ecosystem Balance:** Biodiversity ensures that different species fulfill distinct roles (producers, consumers, decomposers), forming complex food webs and ecological relationships that maintain planetary balance.

4. What is systematic scientific investigation? What are its key steps?

Answer:

- a. Systematic investigation is the structured process scientists use to explore questions, conduct experiments, and discover new facts. Key Steps of Scientific Investigation:
Asking a Focused Question: Identify a specific phenomenon to explore (e.g., "What factors affect how a puri puffs up?").
- b. Identifying Variables:
 - i. **Controllable Variables (Independent Variables):** Factors you can deliberately change or keep constant (e.g., dough thickness, flour type, oil temperature, method of dropping) .
 - ii. **Observable/Measurable Variables (Dependent Variables):** Factors you observe or measure to assess the impact (e.g., whether it puffs [Yes/No], time taken to puff in seconds, thickness of layers) .
- c. **Fair Testing (Controlling Variables):** Change only one variable at a time while keeping all other conditions identical.
- d. **Careful Observation and Record-Keeping:** Record all sensory details, data, and measurements systematically during the experiment.
- e. **Drawing Conclusions & Iterative Questioning:** Use observations to refine understanding and generate new follow-up questions .